

Technical Requirements Document (TRD)

Eila Eco Pencils E-Commerce Platform

1. Purpose

Defines the technical architecture, implementation standards, integrations, database design principles, deployment strategy, and engineering decisions for the Eila Eco Pencils platform.

Engineering Notes

This section should contain detailed implementation decisions, sequence diagrams, data flow, API contracts, configuration examples, deployment considerations, testing evidence, security reviews, trade-offs, scalability discussions, and operational guidelines. It acts as the engineering reference throughout development and maintenance. This section should contain detailed implementation decisions, sequence diagrams, data flow, API contracts, configuration examples, deployment considerations, testing evidence, security reviews, trade-offs, scalability discussions, and operational guidelines. It acts as the engineering reference throughout development and maintenance. This section should contain detailed implementation decisions, sequence diagrams, data flow, API contracts, configuration examples, deployment considerations, testing evidence, security reviews, trade-offs, scalability discussions, and operational guidelines. It acts as the engineering reference throughout development and maintenance.

2. Technology Stack

Frontend: React + Vite + Tailwind CSS. Backend: Node.js + Express (payment verification only). Database/Auth/Storage: Supabase. Payments: Razorpay. Deployment: Vercel + Render.

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3. High-Level Architecture

React communicates with Supabase for data and authentication. Razorpay payment requests are created and verified by the Node.js backend. Supabase stores products, orders, carts, inquiries, and admin profiles.

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4. Frontend Architecture

Component-based React application with pages, reusable UI components, hooks, services, route protection, responsive layouts, loading states, and error boundaries.

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5. Backend Architecture

Express API exposes payment creation and verification endpoints, validates signatures, updates stock, writes orders, and logs failures.

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6. Database Design

Normalized PostgreSQL schema using Supabase. UUID primary keys, foreign keys, indexes, timestamps, RLS policies, audit logs, and soft-delete where appropriate.

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7. Authentication & Authorization

Supabase Auth for admins, role-based authorization via profiles table, protected admin routes, secure session handling.

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8. Storage

Product images stored in Supabase Storage with public URLs, validation, and naming conventions.

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9. Payment Flow

Create Razorpay order → Complete payment → Verify signature → Store payment → Create order
→ Deduct inventory → Redirect to success and WhatsApp.

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10. Security

Environment variables, HTTPS, input validation, RLS, least-privilege policies, backend verification, sanitization, secure secrets handling.

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11. API Specification

Endpoints for payment creation, payment verification, analytics aggregation, and health checks. Supabase handles CRUD operations through client SDK with RLS.

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12. Error Handling

Consistent API responses, user-friendly messages, retry strategies, logging, fallback UI, and monitoring.

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13. Performance

Pagination, lazy image loading, code splitting, caching, optimized queries, indexed database fields, compressed assets.

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14. SEO

Semantic HTML, meta tags, sitemap, robots.txt, Open Graph tags, structured data, fast page load.

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15. Testing Strategy

Unit testing, integration testing, payment testing, responsive testing, browser compatibility, UAT, deployment verification.

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16. Deployment

Frontend on Vercel, backend on Render, Supabase cloud database, environment variables, CI-ready structure.

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17. Monitoring & Logging

Application logs, payment logs, error monitoring, database monitoring, uptime checks, analytics dashboard.

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18. Coding Standards

ESLint, Prettier, reusable components, descriptive naming, modular architecture, documentation and comments.

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19. Future Scalability

Coupons, wishlist, reviews, invoices, multilingual support, mobile app, ERP integration, AI recommendations.

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20. Technical Acceptance Criteria

Secure payments, successful CRUD operations, responsive UI, protected admin routes, RLS enabled, Lighthouse score >90, production deployment readiness.

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